

SHOPSPHERE

E-Commerce Intelligence Business Analytics

End-to-End Project Report, Explanation Guide and Interview Preparation

Project Scope

An end-to-end business analytics project covering business understanding, requirements, data profiling, cleaning, business analysis, Power BI reporting, and sales forecasting.

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Purpose: portfolio, resume support, interview explanation and project understanding

Status at 23 September 2026

1. Executive Summary

ShopSphere is a simulated e-commerce analytics engagement designed to show how a data analyst can take a business problem and turn it into a usable analytical solution. The project does not stop at charts. It documents the business questions, data requirements, data quality work, analytical rules, KPI definitions, business analysis, Power BI model, dashboard design, and a first predictive sales-forecasting workflow.

The core order-level analytical table contains 99,441 orders and 34 fields. After applying the project's delivered-order logic, 96,478 delivered orders were used for the main realized-sales KPIs. The project reports 15.42M in delivered sales, a 97.02% delivery rate, a 3.00% repeat-customer rate, a 6.77% late-delivery rate, and an overall average review score of 4.09.

The predictive phase converts the order-level sales history into a monthly time series, compares a naive baseline with Simple Exponential Smoothing and Holt's linear trend, and validates the models on a time-based holdout period. Simple Exponential Smoothing achieved an 8.66% MAPE on the selected four-month test period and was used for a six-month historical forecast from September 2018 through February 2019.

Important scope note

The forecasting data ends in August 2018. Therefore, the six-month forecast is a historical forecasting exercise, not a current 2026 business forecast.

2. The Project in One Sentence

Interview one-liner

I built an end-to-end e-commerce analytics solution that transformed raw order, customer, product, review, delivery and marketing data into validated business KPIs and 10 Power BI pages, then added a Python-based sales forecasting layer validated with MAE, RMSE and MAPE.

3. Business Problem

ShopSphere has many operational and commercial data points, but raw data alone does not tell management what is happening or where attention is required. The project therefore turns business questions into measurable analysis.

- How much is the business selling and how does monthly sales performance change?
- How many customers return after their first purchase?
- Which categories and locations contribute more sales and customer value?
- How often are orders late and how does delivery experience relate to review scores?
- Which lead sources and lead types convert into closed deals?
- Can historical sales data be used to estimate future sales?

4. Objectives

1. Understand the business context and define measurable analytical questions.
2. Document stakeholders, business rules, functional requirements and non-functional requirements.
3. Identify required entities, fields, relationships and KPIs.
4. Profile and clean the source data.
5. Create reliable analysis-ready tables and a consolidated order-level master table.
6. Perform business analysis across sales, customers, products, operations, reviews, marketing and geography.
7. Build an interactive Power BI report with 10 pages.

8. Validate the key dashboard metrics.
9. Build and evaluate simple sales forecasting models using a time-based validation process.
10. Export predictive results to CSV and connect them to Power BI.

5. End-to-End Lifecycle

```
01 Business Understanding
    ->
02 Business Requirement Gathering
    ->
03 Data Requirement Analysis
    ->
04 Data Collection Strategy
    ->
05 Data Profiling
    ->
06 Data Cleaning
    ->
07 Business Analysis
    ->
08 Power BI Dashboards
    ->
09 Predictive Analytics
    ->
10 Power BI Predictive Insights
```

6. Data and Data Architecture

The project uses multiple e-commerce entities and consolidates them into analysis-ready structures.

Area	Examples
Orders	orders, order status, timestamps, delivery dates
Order items	item count, products, sellers, product sales, freight
Customers	customer IDs, unique customer IDs, city, state
Products	product information and category mapping
Payments	payment records, value, types, installments
Reviews	review score, review presence and aggregated review metrics
Marketing	MQLs, lead origin, lead type, business segment, closed deals
Geography	customer city and state

The core order-level analytical table is Fact_Orders, which is based on master_orders.csv. Supporting tables and dimensions include Customer_Summary, Category_Sales, Category_Experience, Marketing_MQL, Closed_Deals, Dim_Date, Dim_Category, and a centralized _Measures table in Power BI.

7. Data Profiling and Cleaning

The project includes a dedicated profiling phase before business analysis. The purpose is to understand data types, missing values, duplicate identifiers, distributions and structural relationships before using the data for KPIs.

- Order-level uniqueness was checked and the master order table contains 99,441 rows with zero duplicate order IDs.

- Date columns were standardized and converted into usable date/time types.
- Missing values were assessed before aggregation.
- Aggregated payment, item and review fields were created at order level so different source tables could be analyzed consistently.
- Boolean business flags such as is_delivered, is_late, has_review and has_payment were used to make filtering and KPI logic explicit.

Why this matters

A dashboard is only as reliable as its underlying business rules. The project therefore validates the data and KPI logic before using the visuals for business interpretation.

8. Validated Business KPIs

KPI	Value	Simple meaning
Total Orders	99,441	All orders in the master order table
Delivered Orders	96,478	Orders marked as delivered
Delivery Rate	97.02%	Delivered orders divided by total orders
Unique Delivered Customers	93,358	Distinct customers with delivered orders
Repeat Customers	2,801	Customers with at least two delivered orders
Repeat Customer Rate	3.00%	Repeat customers divided by unique delivered customers
Delivered Sales	INR 15,419,773.75	Sales value including freight for delivered orders
Average Order Value	INR 159.83	Delivered sales divided by delivered orders
Average Review Score	4.09	Review-score weighted order-level aggregate
Late Orders	6,534	Delivered orders flagged as late
Late Delivery Rate	6.77%	Late orders divided by delivered orders
Credit Card Penetration	76.94%	Share of orders with credit card payment
Product Sales	INR 13,221,498.11	Product value for delivered orders
Freight Value	INR 2,198,275.64	Freight value for delivered orders
Freight Burden	16.63%	Freight divided by product sales
MQLs	8,000	Marketing-qualified leads
Closed Deals	842	Closed marketing deals
MQL-to-Closed Conversion	10.53%	Closed deals divided by MQLs

9. Key Business Insights

9.1 Customer retention

Repeat customers account for about 3.00% of unique delivered customers. The customer base is therefore dominated by one-time purchasers in this dataset.

Among repeat customers, 2,573 customers placed exactly two orders, which represents about 91.86% of repeat customers. This shows that repeat behavior is concentrated around a second purchase rather than a large population of highly frequent buyers.

9.2 Customer value

Repeat customers have higher average sales per customer than one-time customers. In the project analysis, repeat customers generated average sales of about INR 308.53 per customer, while one-time customers averaged about INR 160.73.

9.3 Delivery and customer satisfaction

The project calculated an on-time order review score of approximately 4.29 and a late order review score of approximately 2.27. This is a descriptive relationship in the analyzed data: late orders are associated with substantially lower average review scores.

9.4 Freight economics

Freight represents approximately 16.63% of product sales overall. Category-level burden varies widely, so a category that appears strong on sales alone may have a less attractive logistics profile.

9.5 Geographic concentration

The top five cities contribute about 27.30% of delivered sales, the top ten contribute about 33.29%, and the top twenty contribute about 40.01%. This shows that geographic performance is not evenly distributed.

9.6 Marketing funnel

The project records 8,000 MQLs and 842 closed deals, giving a 10.53% overall MQL-to-closed conversion rate. Conversion varies by origin, lead type and other lead attributes.

10. Power BI Dashboard: 10 Pages

Page	Purpose	What to say in an interview
1. KPI Validation	Validate the main business numbers	I created a dedicated validation page so core KPI logic could be checked before being presented.
2. Executive Overview	High-level business performance	This is the management entry point for sales, orders, customers, delivery and geography.
3. Sales & Revenue	Sales trends, AOV and revenue drivers	I connected sales value, order volume, AOV, product sales and freight.
4. Product & Category	Category sales, freight, reviews and delivery	I compared category performance from both financial and operational perspectives.
5. Customer Intelligence	One-time vs repeat behavior and customer value	I focused on retention, frequency and customer contribution rather than an overly complex cohort page.
6. Operations & Delivery	Delivery time, lateness and variance	I separated on-time and late delivery performance and examined geographic differences.

7. Customer Satisfaction	Review performance and delivery relationship	I compared customer review outcomes for on-time and late orders.
8. Marketing & Sales Funnel	MQLs, closed deals and conversion	I analyzed the movement from leads to closed deals and differences across lead attributes.
9. Geographic Customer Intelligence	State-level customer and sales value	I compared sales, customers, sales per customer and retention by state.
10. Predictive Insights	Historical sales vs six-month forecast	I integrated the Python forecast output into Power BI so predictive results are visible to business users.

11. Power BI Data Model

The report uses a structured analytical model with a central fact table and supporting dimensions/tables.

Component	Role
Fact_Orders	One row per order; central business event table
Dim_Date	Date filtering and time-based analysis
Dim_Category	Shared category dimension
Category_Sales	Category-level sales and logistics measures
Category_Experience	Category-level review and delivery measures
Customer_Summary	One row per customer with frequency and value attributes
Marketing_MQL	Marketing-qualified leads
Closed_Deals	Closed marketing deals
Funnel_Stages	Funnel labels for the MQL-to-Closed visual
_Measures	Centralized DAX measures

12. Predictive Analytics: Sales Forecasting

The predictive phase asks a new business question: can historical monthly delivered sales be used to estimate future sales?

12.1 Target

Target variable: delivered_sales, calculated from sales_value_incl_freight for delivered orders.

12.2 Monthly preparation

The 99,441-order master table was filtered to 96,478 delivered orders and aggregated by month. The monthly table contains delivered sales, delivered orders, product sales, freight value and AOV.

November 2016 was missing from the delivered monthly records, so a continuous monthly time index was created and November 2016 was represented as zero delivered sales.

12.3 Train-test split

Training period: September 2016 to April 2018.

Test period: May 2018 to August 2018.

The split is chronological rather than random because forecasting must respect time order: the model learns from earlier periods and is tested on later periods it did not see.

13. Forecasting Models and Validation

Three methods were tested on the same holdout period.

Model	MAE	RMSE	MAPE
Simple Exponential Smoothing	INR 87,270.26	INR 100,235.80	8.66%
Naive Baseline	INR 94,366.00	INR 108,854.12	9.37%
Holt Linear Trend	INR 331,448.01	INR 354,242.37	32.57%

Simple Exponential Smoothing had the lowest error on this selected validation period. Holt's trend model performed worse because continuing the earlier upward trend did not match the decline seen in the final test months.

How to explain model selection

I did not choose the most complex model. I compared simple approaches on unseen historical months and used the validation results to choose the final forecasting approach.

14. Final Six-Month Forecast

Month	Forecast Sales
September 2018	INR 995,710.73
October 2018	INR 995,710.73
November 2018	INR 995,710.73
December 2018	INR 995,710.73
January 2019	INR 995,710.73
February 2019	INR 995,710.73

Total six-month forecast: approximately INR 5.97M.

The forecast is flat because the selected Simple Exponential Smoothing configuration estimates the current level without explicitly modeling a trend or seasonal pattern.

15. Python to Power BI Integration

```

master_orders.csv
->
Monthly delivered-sales table
->
Naive / SES / Holt validation
->
Final SES forecast
->
Data/Analysis/sales_forecast.csv
->
Power BI Predictive Insights page

```

- Notebook: 09_Predictive_Analytics/notebooks/01_sales_forecasting.ipynb
- Forecast output: Data/Analysis/sales_forecast.csv
- Model comparison output: 09_Predictive_Analytics/outputs/forecast_model_comparison.csv
- Forecast image: 09_Predictive_Analytics/outputs/sales_forecast.png

16. Reproducing the Predictive Analysis

1. Open the project folder: E:\ShopSphere_Analysis
2. Open 09_Predictive_Analytics/notebooks/01_sales_forecasting.ipynb
3. Confirm Data/Analysis/master_orders.csv exists
4. Install dependencies with `pip install -r requirements.txt`
5. Run the notebook cells in order
6. Confirm Data/Analysis/sales_forecast.csv is created

7. Open the Power BI report and refresh the forecast table if needed

Recommended project dependency file:

```
pandas
numpy
matplotlib
scikit-learn
statsmodels
jupyter
ipykernel
```

17. Limitations and Responsible Interpretation

- The source forecasting history ends in August 2018, so the forecast is historical and should not be presented as a 2026 forecast.
- Only about two years of monthly history are available, which limits confidence in long-term trends and seasonal patterns.
- The final SES configuration does not explicitly model seasonality.
- The test period contains only four months, so the validation result should be treated as evidence for this exercise, not a guarantee of future accuracy.
- External variables such as promotions, price changes, inventory constraints, holidays and competitor actions are not modeled.
- A production forecasting solution should use fresher data and stronger rolling or walk-forward validation.

18. Resume-Ready Project Description

Project title

ShopSphere - E-Commerce Intelligence Business Analytics

Built an end-to-end e-commerce analytics solution using Python, Power BI, Power Query and DAX to profile and clean 99K+ orders, validate KPIs, analyze sales, customer retention, category performance, delivery, satisfaction, marketing funnel and geography, and integrate a validated sales forecast into Power BI.

Resume bullets

- Built an order-level analytical model for 99,441 e-commerce orders and validated key KPIs including 96,478 delivered orders, INR 15.42M delivered sales and 3.00% repeat-customer rate.
- Developed a 10-page Power BI report covering executive performance, sales, category, customer, delivery, satisfaction, marketing and geographic intelligence.
- Analyzed delivery and customer experience, finding an average review score of 2.27 for late orders versus 4.29 for on-time orders in the analyzed dataset.
- Built a Python time-series forecasting workflow comparing Naive Baseline, Simple Exponential Smoothing and Holt Linear Trend using MAE, RMSE and MAPE.
- Integrated the six-month SES sales forecast into Power BI, with 8.66% MAPE on the selected four-month historical holdout period.

Resume accuracy note: add SQL to the project skill line only if SQL analysis is actually included in the repository. The current repository structure inspected on 23 September 2026 does not show a dedicated SQL analysis folder or SQL files.

19. Interview Explanation Guide

19.1 30-second answer

ShopSphere is my end-to-end e-commerce analytics project. I started with business requirements, then profiled and cleaned the source data and built an order-level analytical model. I used Power BI to build 10 pages covering sales, customers, categories, delivery, satisfaction, marketing and geography. I also added a sales forecasting phase in Python, compared three simple forecasting approaches using time-based validation, and connected the final forecast back to Power BI.

19.2 60-second answer

The main business problem was turning fragmented e-commerce data into reliable decision support. I documented the business questions and requirements first, then profiled and cleaned the data and created a 99,441-order master table. From there I calculated validated KPIs such as delivered sales, AOV, repeat-customer rate, late-delivery rate and review performance. I built a structured Power BI model and 10 dashboard pages. For predictive analytics, I aggregated delivered sales monthly, held out the last four months for testing, compared a naive baseline with Simple Exponential Smoothing and Holt, and selected SES because it had the lowest validation error among the tested methods. The final six-month historical forecast was exported as CSV and integrated into Power BI.

19.3 Two-minute answer

I treated the project like a real analytics engagement rather than jumping directly into visualization. The first stage defined the business objective and stakeholders. The next stages defined the required data entities, fields, relationships and KPI rules. I then profiled the source data for missing values, duplicates, types and data quality issues before building an analysis-ready order-level master table. The business analysis covered sales and revenue, customer retention, category economics, operations and delivery, customer satisfaction, marketing funnel performance and geographic concentration. In Power BI I used Fact_Orders as the central fact table with dimensions such as date and category, plus customer and marketing analytical tables. I created 10 report pages, including a KPI validation page to check the main numbers. The predictive extension used the delivered-sales series from the master table. I created a continuous monthly time series, used a chronological train-test split, evaluated a naive baseline, Simple Exponential Smoothing and Holt's trend method, and compared MAE, RMSE and MAPE. SES achieved 8.66% MAPE on the four-month holdout and was used for the final six-month historical forecast. The prediction output is stored in CSV format and consumed by the Power BI predictive page. The main limitation is that the source history ends in 2018, so the forecast demonstrates the workflow but is not a current 2026 forecast.

20. Common Interview Questions and Simple Answers

Q: Why did you create Fact_Orders?

A: It gives one row per order, which makes order-level KPIs consistent and gives the report a clean analytical base.

Q: Why use delivered orders for realized sales?

A: The project uses delivered orders to represent completed sales consistently with the main business analysis.

Q: How did you calculate repeat customers?

A: A customer is treated as repeat when the customer has at least two delivered orders.

Q: Why is repeat-customer rate only 3.00%?

A: Because the delivered customer base is heavily dominated by customers with only one delivered order in this dataset.

Q: Why did you use a time-based train-test split?

A: Because future data should not be mixed into model training in a forecasting problem.

Q: Why did you test a naive baseline?

A: It provides a simple reference. A forecasting model should show improvement over a basic benchmark before being presented as useful.

Q: Why did you test Holt?

A: Holt models both level and trend, so it is a natural comparison when the historical series shows growth.

Q: Why did Holt perform poorly?

A: The model continued the earlier trend, but the held-out months were lower than the recent peak, so the trend extrapolation created larger errors.

Q: Why is the SES forecast flat?

A: The selected SES configuration estimates the current level without an explicit trend or seasonal component, so future values remain at a similar level.

Q: What does MAPE 8.66% mean?

A: On the selected four-month historical holdout, the average absolute percentage difference between actual and SES forecast was about 8.66%.

Q: How does Python connect to Power BI?

A: Python writes the validated forecast results to CSV; Power BI imports the CSV and displays the forecast alongside the historical data.

Q: What would you do next?

A: Use fresher data, rolling validation, richer seasonal models, and potentially add delivery-risk, repeat-purchase or category-level forecasting models.

21. Final Repository QA - Important Before GitHub Finalization

The local project screenshots supplied on 23 September 2026 show the new predictive folder, forecast outputs, Page 10 screenshot and a local dependency file. However, the connected GitHub repository was also checked directly.

Item	Local workspace screenshot	GitHub main branch check
09_Predictive_Analytics/	Present	Not present yet
01-08 project folders	Present	Present
reports 1-9	Present	Present
reports Page 10 screenshot	Present, but filename ends with .png.png	Not present yet
Data/Analysis/master_orders.csv	Present	Present

Data/Analysis/sales_forecast.csv	Present	Not present yet
forecast_model_comparison.csv	Present	Not present yet
sales_forecast.png	Present	Not present yet
Root README.md	Updated locally	Still old on GitHub
Dependency file	Local file is named requirement.txt	No dependency file found
What remains to sync The analytical work is complete, but the current GitHub main branch is behind the local workspace. Final packaging should include the 09_Predictive_Analytics folder, sales_forecast.csv, Page 10 screenshot, a correctly named requirements.txt, the updated root README.md, and then one final git add/commit/push.		

22. Small Structure Fixes

- Rename requirement.txt to requirements.txt. This is the conventional project filename and is easier for recruiters and collaborators to recognize.
- Rename the Page 10 screenshot from '10. Predictive Insights - Sales Forecast.png.png' to '10. Predictive Insights - Sales Forecast.png' to remove the duplicated .png extension.
- Keep folder naming consistent. Your local screenshot shows '05_Data_profiling' while the GitHub repository currently uses '05_Data_Profiling'. On Windows this can be hard to notice because the filesystem is case-insensitive; keep one spelling everywhere.
- The Data/Staging folder is ignored by .gitignore and can remain local if it is intentionally temporary. Git does not track an empty directory by itself.

23. Conclusion

ShopSphere now demonstrates a complete analytical story: a business problem is converted into requirements, data is profiled and cleaned, business rules are made explicit, KPIs are validated, insights are delivered through Power BI, and a predictive layer is added with model comparison and validation.

The strongest way to present this project in an interview is not to memorize every number. Instead, explain the logic: why the business problem matters, how the data was prepared, how the KPI definitions were chosen, why the dashboard model is structured the way it is, how the forecasting models were compared, and what the limitations are.

Final message to remember

The project is about turning data into reliable business decisions - first by explaining what happened, then by adding a carefully validated estimate of what may happen next.

Appendix A - Final Local Project Structure

```
ShopSphere_Analysis/
├── 01_Business_Understanding/
├── 02_Business_Requirement_Gathering/
├── 03_Data_Requirement_Analysis/
├── 04_Data_Collection_strategy/
├── 05_Data_Profiling/
├── 06_Data_Cleaning/
├── 07_Business_Analysis/
├── 08_Dashboards/
│   ├── README.md
│   └── ShopSphere_Business_Intelligence.pbix
├── 09_Predictive_Analytics/
│   ├── notebooks/
│   │   └── 01_sales_forecasting.ipynb
│   └── outputs/
```

```

├── forecast_model_comparison.csv
├── sales_forecast.png
├── README.md
├── Data/
│   ├── Raw/
│   ├── Cleaned/
│   └── Analysis/
│       ├── master_orders.csv
│       └── sales_forecast.csv
├── reports/
│   ├── 1. KPI Validation.png
│   ├── 2. Executive Overview.png
│   ├── 3. Sales & Revenue.png
│   ├── 4. Product & Category.png
│   ├── 5. Customer Intelligence.png
│   ├── 6. Operations & Delivery.png
│   ├── 7. Customer Satisfaction.png
│   ├── 8. Marketing & Sales Funnel.png
│   ├── 9. Geographic Customer Intelligence.png
│   └── 10. Predictive Insights - Sales Forecast.png
├── .gitignore
├── requirements.txt
└── README.md

```

Appendix B - Reference Screenshots

Predictive Insights dashboard snapshot:

